



Risk Estimation of Preeclampsia

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Introduction

Preeclampsia is a hypertensive disorder that may occur after 20 weeks of pregnancy. It can cause liver damage, low platelet count, preterm birth, etc. If left unaddressed, the disorder can rapidly worsen to Eclampsia, meaning it will affect brain function, putting the mother at risk of comas or seizures. As there is no immediate treatment for Preeclampsia outside of childbirth, it is imperative to find a way to predict a mother's risk of the disorder so steps can be taken to prevent it. To do this, we use different methods of machine learning on 12-Lead ECGs to accurately predict risk.

Future Work

- **Lead Reduction:** Instead of using 12 leads of ECG data, a reduction in amount of leads may be enough in detecting preeclampsia.
- **Feature Importance Across Models:** Feature importance selection in this project were results of a Random Forest model, though models such as CatBoost or LightGBM may find other features removed useful. This has the possibility of improving performance metrics at the cost of more features.
- **MUSE derived features:** During NeuroKit validation in the earlier stages of the project, instead of deriving novel features from the raw ECGs, it was possible to derive comparable features using only MUSE-reported data. Current models used MUSE + extracted NeuroKit features, but what if derived MUSE features were used instead?

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CNN Models for ECG Feature Learning

Decision-Tree Methods

UC Davis Health was able to provide an unbalanced dataset of ECGs with already reported features exported from a GE MUSE 12L system. This dataset contains 329 preeclamptic ECGs and 1830 normal ECGs for a split of 15.2%/84.8%.

1. Preprocess ECG data using NeuroKit2 to extract more features
2. Add extracted NeuroKit2 features to the MUSE reported dataset
3. Apply feature importance methods
 - Pearson Correlation
 - ANOVA
 - Recursive Feature Elimination
4. Train decision-tree models

Original feature importance and modeling was performed on a Random Forest model using 5-fold StratifiedGroupKFold and RandomUnderSampler. Eventually, testing of other decision-tree models and ensembles were performed to obtain better results.

Decision-Tree Results

Unweighted Voting with SVM and LogisticRegression proved to be the best model when focusing on sensitivity/balanced accuracy. We look at these metrics to understand how well the model is correctly identifying preeclamptic ECGs and whether or not the model is genuinely distinguishing correctly between a normal or preeclamptic ECG.

Ensemble	AUROC	AUPRC	Specificity	Sensitivity	Balanced Accuracy	Precision	F1-Score
Unweighted Voting	74.54 ± 4.16	41.81 ± 7.86	74.26 ± 3.95	59.32 ± 10.70	66.79 ± 3.60	29.25 ± 1.38	38.94 ± 2.99
Weighted Voting	74.75 ± 4.48	42.26 ± 7.92	74.75 ± 4.48	60.23 ± 10.88	67.49 ± 3.45	30.05 ± 1.57	39.80 ± 2.76
Unweighted Voting + SVM, LR	75.72 ± 3.94	41.58 ± 8.03	74.64 ± 4.70	62.35 ± 10.93	68.50 ± 3.22	30.72 ± 1.08	40.85 ± 2.24
Stacking	74.77 ± 4.35	42.43 ± 7.59	74.37 ± 4.18	59.62 ± 10.80	67.00 ± 3.46	29.45 ± 1.20	39.17 ± 2.79
Bagging CatBoost	74.81 ± 4.98	41.96 ± 8.86	74.70 ± 4.68	59.62 ± 15.19	67.16 ± 5.59	29.49 ± 2.51	39.10 ± 5.21

Table 1. Performance comparison of tree-based ensemble methods on the UCDH dataset.

To identify the most robust cardiological markers for preeclampsia and eliminate noisy variables, we evaluated our engineered feature space across multiple selection feature importances: Pearson Correlation, ANOVA, and RFE (scoring on balanced accuracy, rocauc, and average precision).

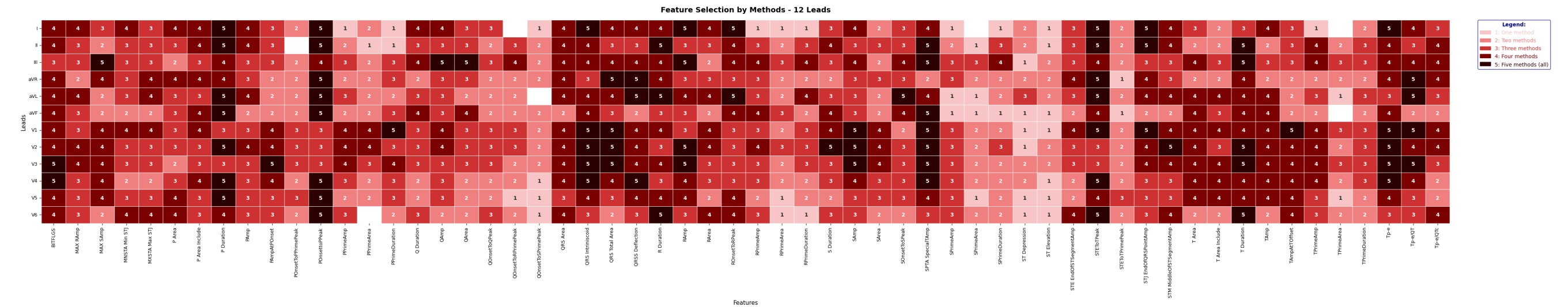


Figure 1. Feature Selection consensus heatmap across 12 ECG leads.

The horizontal distribution illustrates consensus across feature importances, where darker blocks reveal important features for the lead. Global features are not reported here.